Pratik Relekar

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Research Experience

Data Science Research Services (DSRS), UIUC, Chicago, USA **Sep 2022 – Dec 2024**

Data Scientist (Research)

- Supported accountancy research on stock market reactions via financial statement sentiment analysis using BERT and advanced NLP models, achieving 83% classification accuracy.
- Constructed a customized OCR pipeline to convert 2,000+ tax documents and 10-K filings into machine-readable text for shareholder perception research.
- Engineered a longitudinal data pipeline aggregating 12 years (2011–2022) of Illinois school and district-level records from EdFacts, CCD, and ISBE sources, harmonizing heterogeneous file formats and converting to Stata datasets for econometric analysis.
- Investigated county-level relationships between educational attainment and cannabis usage across Illinois households, constructing panel datasets linking socioeconomic and education indicators to substance use patterns to establish causal inference.

Experience

Johnson Controls Inc. Chicago, USA **Jan 2025 – Present**

Data Scientist

- Architected a Customer Master hierarchy in Snowflake, developing a parent-child mapping algorithm combining LLM embeddings matching with Snowflake Cortex AI to link dealers and subsidiaries to parent companies at 90%+ accuracy, enabling granular market share analysis for identification of top 20 accounts driving 80% of Global Products revenue.
- Designed and benchmarked a multi-stage entity resolution system for 1.7M+ global customer addresses, evaluating probabilistic matching, transformer-based embeddings against a hand-labeled validation set; selected Azure Maps after rigorous accuracy/cost trade-off analysis, achieving 94.6% clean data rate and eliminating $60K/year in manual review costs.
- Developed a confidence-threshold classifier to flag the residual 5.3% of low-confidence predictions for active learning review, reducing manual validation effort by 400 person-days annually and enabling iterative model improvement on edge-case records.
- Automated the end-to-end Point of Shipment data loading process using Power Automate and Copilot, engineering complex data transformations across multiple business units with deep institutional knowledge and business expertise to replace manual dealer data collection, saving 192 hours/month.

Johnson Controls International PLC, USA **Nov 2021 – Jul 2022**

Business Analyst

- Boosted sales lead identification by 30% by deploying Named Entity Recognition with fuzzy matching to deduplicate and cleanse 1M+ CRM records.
- Developed an ETL pipeline using SQL stored procedures to normalize data according to regional and departmental standards, improving user query speed by 40%.
- Optimized SQL queries powering Power BI dashboards for sales, reducing refresh times by 20%.

Indium Software Pvt Ltd **Jun 2021 – Oct 2021**

Data Analyst/ Business Intelligence Analyst

- Boosted client engagement and conversion rates by 15% by designing 50+ Tableau dashboards delivering actionable customer insights and accelerated data-driven decision-making.

Skymet Weather Services Ltd (Dept. of Agriculture, Maharashtra) **Nov 2019 – May 2021**

Data Analyst

- Slashed query time by 75% by migrating agricultural data from flat files to PostgreSQL and building a streamlined front-end interface for faster data retrieval.
- Implemented machine learning algorithms including Linear Regression, KNN regression, Random Forest, and ANN to predict agricultural production, informing design of crop-advisory policies, forecasting loan defaulters across 25+ districts.
- Engineered a soil health classification model using ensemble methods (Random Forest, Boosting) achieving 85% accuracy, enabling data-driven soil management strategies.

Bharat Forge Ltd & Trimurti Stampings Pvt Ltd **Jan 2015 – Oct 2019**

Design Engineer/Electrical Engineer

Open Source Contribution

NielsenIQRetail | *Python · PyPI · Open Source* **Aug 2024**

- Pioneered the first-of-its-kind open-source Python library for parallel reading and processing of NielsenIQ Retail Scanner Data, published on PyPI and endorsed by NielsenIQ and Chicago Booth School of Business.

Skills

Programming: Python (Pandas, Polars, NumPy, scikit-learn, PyTorch, NLTK, spaCy), SQL

Data Engineering Tools: Snowflake, PySpark, ETL, PostgreSQL, MySQL, AWS, Dask

Machine Learning & AI: Natural Language Processing, Machine Learning (ML), Neural Networks, Text Classification

NLP & Document Intelligence: NLTK, spaCy, BERT-based modeling, OCR pipelines, Unstructured Text Analysis, LLMs

Data Visualization Skills: Tableau, Microsoft Power BI, Python (Seaborn, Matplotlib, ggplot)

Education

University of Illinois at Urbana Champaign (UIUC), USA **Aug 2022 – May 2023**

Master of Science in Business Analytics (MSBA) | ***GPA: 3.85/4*** | ***Top 10%***

University of Pune, India

Bachelor of Engineering in Electrical | ***First class with Distinction***